Monil Arora

Vancouver, BC

(+1) 236-777-5739 monilarora2003@gmail.com linkedin portfolio

Education

University of British Columbia, Vancouver **Expected 2027**

Bachelor of Applied Science in Electrical Engineering

Relevant Coursework: Digital System Design, Computing Systems, Signals and Systems, Systems and Controls

Work Experience

Laser Zentrum Hannover (LZH) **Jan – Jul 2026**

Research Intern *Hannover, Germany*

- Extended **STM32F401CC** firmware in **C++/FreeRTOS** packetizing telemetry into **CCSDS SPP** frames, **SLIP**-encoded over **RS-422** through a **Lantronix UDS2100** bridge, with **COP-1/FARM** delivery, a **LittleFS** boot counter and RTC time persistence in flash, and a priority **bypass-command queue**; diagnosed and fixed an **IWDG** reset mid image-transfer by relocating task check-ins inside the transmission loop.
- Extended **YAMCS** Mission Control with custom **Java data links** for SLIP-over-UDP with accumulation-buffer reassembly, **COP-1 FOP**-compliant TC delivery, an **XTCE mission database** decoding **APID 130 TM / 100 TC** with threshold alarms and telemetry-based **command verifiers**, and a **yamcs-client** script automating **COP-1** reboot recovery.
- Implemented end-to-end **image downlink** with an 8 KB SRAM buffer sending 79-byte **APID 131 / VCID 1** chunks on firmware, and a Python **WebSocket** reassembler reconstructing JPEGs into YAMCS object storage; validated against a **PyQt6** payload simulator, **Python TCP-UDP bridge**, and a **hardware-in-the-loop Basler GigE** camera.

Sharang Shakti Pvt Ltd **May – Jun 2024**

Embedded Signal Processing Engineer *Gurugram, India*

- Contributed to the **digitalization of an analog radar system** by designing and optimizing high-frequency waveform generation and signal conditioning modules using the **AD9910 DDS**, improving signal resolution and reducing phase noise.
- Integrated the **AD8336 Variable Gain Amplifier (VGA)** with the **AD9910** for real-time amplitude modulation and dynamic range adjustment in **high-frequency radar signals**.

Projects

Self-Balancing Robot | *C, Arduino, ESP, Sensors, Embedded Systems* **Jan – Apr 2025**

- Designed and built a two-wheeled **self-balancing robot** using **Arduino Nano 33**, **IMU sensors**, and **motor drivers**, implementing real-time **PWM control** for tilt stabilization up to **45°**.
- Developed a **BLE-enabled mobile app** and **LCD remote** using **ESP32** to transmit commands and display RPM, tilt angle, and sonar distance readings.
- Validated proportional control through **oscilloscope analysis**, achieving approximately **30% improvement in stability**.

Deep Neural Network Accelerator | *Verilog, Assembly, C, FPGA* **Oct – Nov 2024**

- Implemented a **deep neural network accelerator** on an embedded **Nios II system** using the **DE1-SoC board** for MNIST classification with **Q16.16 fixed-point arithmetic** and **ReLU** activation.
- Developed a **system-on-chip (SoC)** featuring an **SDRAM interface**, applying **Quartus timing constraints** to ensure stable operation at **50 MHz**.

Metal Detector Robot | *C, STM32, EFM8, JDY40 Bluetooth* **Mar – Apr 2024**

- Collaborated with a team of six to design a battery-powered, **remote-controlled metal-detection robot** using **STM32** and **EFM8 microcontrollers**.
- Programmed robust **communication protocols** and an **LCD interface** for real-time visualization of **metal detection intensity** and **magnetic field strength (up to 1.4 T)**.

Reflow Oven Controller | *Assembly, ARM* **Jan – Feb 2024**

- Developed a **microcontroller-based reflow oven controller** using the **N76E00 (ARM architecture)** in **assembly language** to automate **soldering temperature profiles** ranging from **25°C to 240°C**.
- Achieved precision within **±3°C**, integrating **automatic safety shutdowns and safety alerts**, a **real-time PC monitoring interface**, and support for multi-stage temperature ramping.

Technical Skills

Languages: C/C++, Python, Java, Assembly, Verilog
Tools: YAMCS, FreeRTOS, PlatformIO, Maven, Quartus, ModelSim, CrossIDE, Cadence, MATLAB, LTspice, Git
Hardware: STM32, ESP32, EFM8, Arduino, DE1-SoC, Raspberry Pi, IMU Sensors, Motor Drivers